

Azure Cloud Setup for EVIL Airport

Introduction

EVIL Airport is a new startup airport built to handle overflow traffic from LAX and to serve major worldwide customers including shipping partners like FedEx and airlines like Delta. Because the airport supports critical flight and shipping operations, the Azure environment was designed to give the business scalability, high availability, layered security, and a clear disaster recovery path. The hybrid portion of the design is kept in electronic documentation that does not live on the live servers.

Business Requirements

- A scalable backend that can grow with traffic
- A disaster recovery plan across regions
- High availability for global partners
- Strong network and identity security
- Local and remote access based on job role
- Performance and functional testing
- Compliance with current regulatory standards

Server

The existing virtual machine fittrack-server was reused as the central backend server for the EVIL Airport portal. It handles user authentication, flight schedule updates, and the shipping partner intake forms.

Virtual Machine

A separate virtual machine was provisioned to represent the internal operations network. This VM is reachable only through a private route from the main server, which limits the attack surface and keeps internal tools off the public internet.

Security

A network security group was applied to the VMs with rules that allow only authorized traffic. Public access for management ports is closed and only opens for approved admin IPs. Shipping and airline partner traffic is filtered so that only approved partner ranges reach the backend.

Storage

A storage account was created to hold the airport's operational documents such as manifests, schedules, and partner agreements. A container was used to group these files so the staff can find them quickly while keeping access controlled.

Archive Storage

Files that are no longer needed for everyday work were moved into the Archive tier. The Archive tier saves money on storage and is the right fit for records that EVIL Airport still has to retain for regulatory and audit purposes.

Disaster Recovery

Geo redundant storage was turned on for the storage account so the operational data is replicated to a paired Azure region. The hybrid documentation lives outside of the live servers, which means it survives even a full region outage and supports a clean recovery process.

Identity and Access Management

Three identities were created to model the access roles, an EVIL Airport administrator, a FedEx shipping partner user, and a Delta airline partner user. Role based access control was used so each identity can reach only the resources that match its job. Staff get local access through their administrator account and partners get remote access through their partner accounts.

Monitoring and Testing

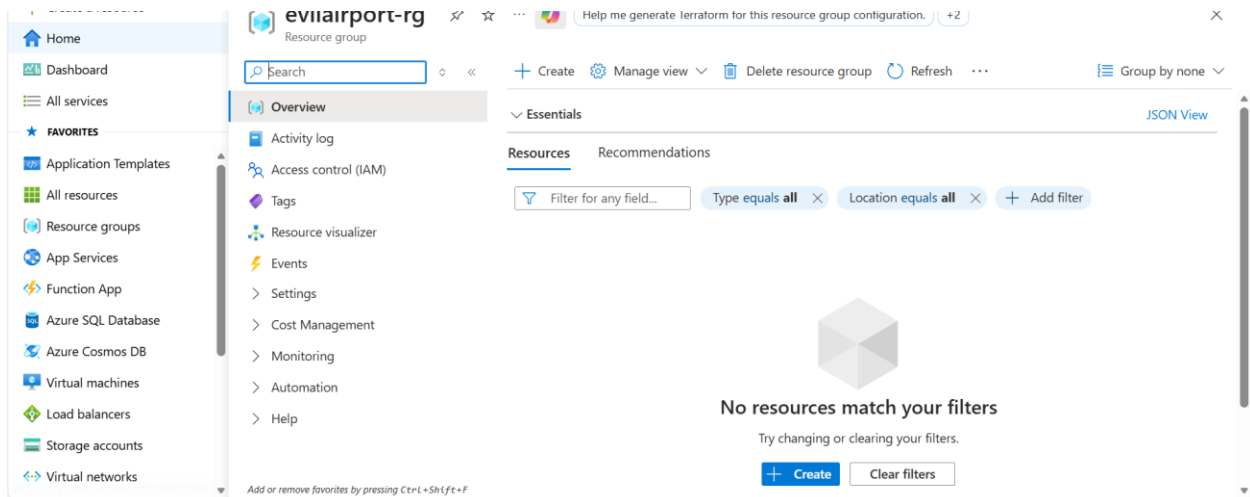
Azure Monitor was set up to track VM uptime, CPU load, and storage latency. Functional testing was done by signing into each role and confirming the access boundaries. Performance testing was run to measure response under simulated partner load and the results were saved for later comparison.

Security and Encryption

Encryption at rest is applied to all Azure storage by default with platform managed keys. Data in transit is sent over TLS. These controls cover the live operational data, partner documents, and the archived records.

Conclusion

The Azure environment for EVIL Airport meets the startup's needs by providing a scalable, available, secure, and recoverable platform that handles flight and shipping operations for partners worldwide.



fittrack-server Virtual machine

Help me copy this VM in any region | Manage this VM with Azure CLI

Advisor (1 of 2): Use Availability zones for better resiliency and availability →

Help me copy this VM in any region

Connect ▾ ▶ Start ↺ Restart □ Stop ⌚ Hibernate 📷 Capture ▾ 🗑 Delete ↻ Refresh ⋮

Essentials

JSON View

Resource group (move)	Operating system
fittrack-rg	Linux (ubuntu 24.04)
Status	Size
Running	Standard B2ts v2 (2 vcpus, 1 GiB memory)
Location	Primary NIC public IP
West US 2	20.114.7.79
Subscription (move)	1 associated public IPs
Azure subscription 1	Virtual network/subnet
Subscription ID	fittrack-server-vnet/default
8068344b-b286-45b3-bae5-79c137f0c580	DNS name
	Not configured
	Health state

https://portal.azure.com/#home

Add or remove favorites by pressing Ctrl+Shift+F

evil-vnet-1778289695167 Overview

Deployment

Delete Cancel Redeploy Download Refresh

✓ Your deployment is complete

Deployment name : evil-vnet-1778289695167

Subscription : Azure subscription 1

Resource group : evilairport-rg

Start time : 5/8/2026, 6:21:38 PM

Correlation ID : 7b730459-89c5-4627-ace5-defaf466c0f3

Deployment details

Next steps

Go to resource

Add or remove favorites by pressing Ctrl+Shift+F

evil-ns3 Inbound security rules

Network security group

+ Add Hide default rules Refresh Delete Give feedback

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

Priority	Name	Port	Protocol	Source
100	Allow-SSH	22	TCP	Any
65000	AllowVnetInBound	Any	Any	VirtualNetwork
65001	AllowAzureLoadBalan...	Any	Any	AzureLoadBalancer
65500	DenyAllInBound	Any	Any	Any

Add or remove favorites by pressing Ctrl+Shift+F

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

evilstorage123

Storage account

Search

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Create an LLM rule for my storage account

Analyze alerts for this storage account

Upload

Open in Explorer

Delete

Move

Refresh

Open in mobile

CLI / PS

Essentials

Resource group (move)

evlairport-rg

Location

westus2

Primary/Secondary Location

Primary: West US 2, Secondary: West Central US

Subscription (move)

Azure subscription 1

Subscription ID

8068344b-b286-45b3-bae5-79c137f0c580

Disk state

Primary: Available, Secondary: Available

Tags (edit)

Add tags

Performance

Standard

Replication

Read-access geo-redundant storage (RA-GRS)

Account kind

StorageV2 (general purpose v2)

Provisioning state

Succeeded

Created

5/8/2026, 6:31:26 PM

JSON View

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

evilstorage123

Storage account

Search

Containers

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

File shares

Queues

Tables

Security + networking

Data management

Add container

Upload

Refresh

Delete

Change access level

Restore containers

Search containers by prefix

Only show active containers

Showing all 2 items

	Name	Last modified	Anonymous access level	Lease state	Storage
☐	\$logs	5/8/2026, 6:31:48 PM	Private	Available	-
☐	airport-files	5/8/2026, 6:35:42 PM	Private	Available	-

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

This blob is currently archived and can't be downloaded. Rehydrate the blob by copying and pasting it in the desired storage account to download or change the blob's tier to hot or cool. [Learn more](#)

Overview

Versions

Snapshots

Edit

Generate SAS

Properties

URL

https://evilstorage123.b...

LAST MODIFIED

5/8/2026, 6:36:57 PM

CREATION TIME

5/8/2026, 6:36:57 PM

VERSION ID

-

TYPE

Block blob

SIZE

237.37 KiB

ACCESS TIER

Archive

ACCESS TIER LAST MODIFIED

5/8/2026, 6:37:14 PM

ARCHIVE STATUS

-

REHYDRATE PRIORITY

-

SERVER ENCRYPTED

true

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Storage Actions

Redundancy

Data protection

Object replication

Blob inventory

Static website

Lifecycle management

Azure AI Search

Settings

Monitoring

Automation

Help

Storage account

Search

Save Discard Prepare for failover Refresh Give feedback

Azure Storage redundancy copies your data so that it is protected from transient hardware failures, network or power outages, and natural disasters. If an outage renders the primary endpoint unavailable, then you can initiate a failover to the secondary endpoint to rapidly restore write access to your data. [Learn more.](#)

Redundancy

Need help with changing redundancy setting? [Help with Copilot](#)

Geo priority replication (Blob only) ☐

Last failover time -

Storage endpoints [View all](#)

Location	Data center type	Status	Failover
West US 2	Primary	Available	-
West Central US	Secondary	Available	-

https://go.microsoft.com/fwlink/?Linkid=2301930

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

All users

Audit logs

Sign-in logs

Diagnose and solve problems

Deleted users

Password reset

User settings

Bulk operation results

Bulk operation results (Preview)

New support request

Default Directory

+ New user Edit Delete Download users (Preview) Bulk operations Refresh

Search users Add filter

6 users found

	Display name ↑	User principal name ↑	User type	Is Agent	On...
☐	DP Delta Partner	delta-user@kidan...	Member	No	No
☐	EO Employee One	employee1@kida...	Member	No	No
☐	ET Employee Two	employee2@kida...	Member	No	No
☐	FP FedEx Partner	fedex-user@kidan...	Member	No	No
☐	LK Loza Kidanemariam	kidanemariamloza...	Member	No	No
☐	SU Staff User	staff-user@kidane...	Member	No	No

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Deployments

Security

Deployment stacks

Policies

Properties

Resource group

Search

+ Add Download role assignments Edit columns Refresh Delete Feedback

Name ↑↓	Type ↑↓	Role ↑↓	Scope ↑↓	Condition ↑↓
Owner (2)				
☐	LK Loza Kidanemariam ae9c8e29-5d49...	User	Owner	Subscription (Inhe... None
☐	LK Loza Kidanemariam ae9c8e29-5d49...	User	Owner	Subscription (Inhe... None
Contributor (1)				
☐	SU Staff User 51bc8218-8502...	User	Contributor	This resource None
Reader (2)				
☐	DP Delta Partner e459a2c1-7a7e...	User	Reader	This resource None
☐	FP FedEx Partner 67584dd8-612...	User	Reader	This resource None

Showing 1 - 5 of 5 results.

https://portal.azure.com/#

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Operations

Auto-shutdown

Run command

Updates

Health monitoring

Configuration management

Policies

Inventory

Change tracking

Virtual machine

Search

Refresh Resource Group Azure Monitor Diagnose And Solve Problems Monitor Settings

Time range: Last 6 hours

Overview Map

VM availability ☐ Available

Azure outages ☐ No outages

Health events ☐ No events

Network Troubleshooter ☐

Run all tests

Checks for common network security rules at RDP, HTTP, HTTPS.

Or more network diagnostic tools, visit [Network Watcher](#)

